Certified branch-and-bound for heterotic line-bundle standard models

Complete slope obstructions, charge-unbounded slices, first-principles equivariant descent, and the holomorphic Yukawa sector

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Abstract

We treat the construction of $E_8 \times E_8$ heterotic compactifications on favourable complete-intersection Calabi–Yau threefolds, with gauge bundle a sum of five line bundles, as an exact search over integer charge matrices in which every pruning decision is a proof and every outcome carries its logical status. The central tool is a complete, coefficient-unbounded form of the slope (poly-stability) obstruction: it holds exactly when the span of the charge vectors meets a fixed polyhedral cone, it is monotone and so prunes every branch of the search tree, and it is decided by a Gordan-type dual test whose cost falls with depth; it removes a spurious candidate retained by the finite test of earlier scans. A slice decomposition with a per-slice anomaly bound turns an explicit family of slices into searches *with no charge cutoff*—425 of 469 slices of the tetraquadric are proven empty for arbitrarily large charges—and an index-divisibility obstruction gives the only fully unbounded exclusions, both confirmed by the published catalog. The engine reproduces the published GUT catalogs of the tetraquadric (10 and 58 models) and of CICY 7447 (93 models) exactly, independent replays exhaust domains of up to 3.5×10^9 states, and run over all 68 favourable CICYs it recovers every published model inside its search boxes. An exact Gröbner and interval-arithmetic decision of stability, itself checked never to reject a published model, classifies the extra survivors and certifies one poly-stable GUT model that the published catalog misses. Computing the Γ-representation on cohomology from the group action (reproducing all 27,821 published character tables exactly), we search rather than verify: 21 orbits on the tetraquadric quotient admit three chiral families, no charged exotics and a Higgs pair, 11 of them absent from the published standard-model database—each because it lies outside that scan's own stated domain. Treating the $U(1)^4$ selection rules exactly, as network-flow membership, we reproduce every operator table of that database for all 2,012 models and find 954 of 2,007 down-type Yukawa tables internally inconsistent. Evaluating leading-order couplings as cohomological triple products, independent of complex structure, and running new and published models through one pipeline, we find that no model on the quotient combines a leading-order top Yukawa, rank-three Yukawa matrices and the absence of dangerous operators; the new models fail for the same reasons, at a similar rate, as the published ones.

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1 Introduction

Heterotic compactifications on smooth Calabi–Yau threefolds with gauge bundles built from sums of line bundles are among the most successful constructions of string vacua with the charged spectrum of the minimal supersymmetric Standard Model (MSSM) [1, 2, 3]. The construction is combinatorial at its core. On a favourable complete-intersection Calabi–Yau (CICY) threefold X with $h := h^{1,1}(X)$, a rank-five bundle $V = \bigoplus_{a=1}^{5} \mathcal{O}_X(k_a)$ is determined by an integer $h \times 5$ charge matrix $K = (k_1, \ldots, k_5)$, and the conditions for a consistent, supersymmetric $SU(5)$ grand unified theory with three chiral families—vanishing first Chern class, the correct chiral index, the Green–Schwarz anomaly condition and poly-stability—are conditions on K and on the topological data of X. Descending to a Standard-Model-like theory then requires a freely acting symmetry Γ, an equivariant structure on V and a Wilson line.

Earlier work turned this into a systematic scan: [3] enumerated all bundles on the 68 favourable CICYs with $h \leq 6$ whose charges lie in a finite box, producing a catalog of 63,325 GUT models, and [1, 2] produced a database of downstairs standard models together with their superpotential selection rules. Machine-learning and heuristic optimisers have since been applied to the same space [4]. These are powerful, but they leave three questions open that this paper addresses.

What does a negative search result prove? A scan over a charge box $|k_a^i| \leq K$ that finds nothing establishes, at best, a statement about that box, and only if every filter used to discard candidates is a genuine *necessary* condition. Several filters in the literature are finite truncations of conditions that quantify over infinite sets (the slope obstruction quantifies over all integer combinations of the charge vectors), and whether their truncations are sound, complete or neither has not, to our knowledge, been settled. Nor is it obvious which parts of the landscape can be excluded *without* a charge cutoff.

Can the downstairs spectrum be computed, not looked up? The Γ-representation content of bundle cohomology decides which states survive the quotient. In existing databases

it is supplied as data. A search that discovers new downstairs models needs to compute it from the group action.

What does the superpotential actually do? Charge conservation under the $U(1)^4$ remnant of the structure group is a necessary condition for a coupling. It is not sufficient: a coupling allowed by the charges can vanish for cohomological reasons. Distinguishing the two needs the actual cup product.

1.1 Summary of results

We build a search engine whose outputs carry their precise logical status and whose every pruning decision is a proof, and use it to answer these questions on the two benchmark geometries of [3], the tetraquadric X_{7862} ($h = 4$) and X_{7447} ($h = 5$). The main results are:

1. **A complete slope obstruction** (Section 4). The finite coefficient test of [3] is sound but incomplete. Its complete form is equivalent to the real span of the charge vectors meeting a fixed rational cone (Proposition 4.3); it is monotone in the columns, so it prunes every branch of the search tree (Proposition 4.5); and it is decided exactly by a Gordan-type dual test whose cost *falls* with depth (Proposition 4.7). On X_{7862} it removes a spurious orbit that the finite test retains at $K = 3$ and $K = 4$, and it is what brings our survivor sets into exact agreement with the published catalog.

2. **Searches with no charge cutoff** (Section 5). Every slope-feasible model lies in a slice $p^\perp$ for a primitive positive integer vector p (Proposition 5.1), and on slices satisfying $(\sum_i p_i)^2 > (h-1)|p|^2$ the anomaly condition bounds every charge (Proposition 5.3). Such slices can therefore be searched exhaustively over *all* integer charges; on X_{7862}, 425 of 469 are proven empty. We isolate exactly where the remaining difficulty lies.

3. **An unbounded arithmetic obstruction** (Proposition 4.12): $d_X := \gcd\{\mathrm{ind}\,\mathcal{O}_X(k)\}$ must divide $3|\Gamma|$. Its two zero-model predictions across the 68 favourable CICYs are both confirmed.

4. **Certified cohomology** (Section 6). A single nonzero $r \times r$ minor modulo any prime certifies the generic rank of a Koszul map in characteristic zero, with no good-reduction hypothesis (Proposition 6.1).

5. **Exact reproduction and independent replay** (Section 7). The engine reproduces the published GUT catalogs of X_{7862} ($\Gamma = \mathbb{Z}_2$: all 10 models; $\Gamma = \mathbb{Z}_2 \times \mathbb{Z}_2$: all 58) and X_{7447} (all 93) model for model; the X_{7447} counts were predicted before the runs. The complete $K = 1$ domains—4.3×10^7 and 3.5×10^9 ordered states—are replayed by an independent implementation with exact survivor-set equality.

6. **All 68 favourable CICYs** (Section 7.3). Run unchanged on every key of the published catalog (271 runs, $h^{1,1} \le 6$, $K \le 3$), the engine never misses a published model inside its boxes. An exact decision of stability (Proposition 4.10) certifies 53 of the 80 extra survivors unstable and finds one poly-stable, anomaly-free, correct-index GUT model on CICY 5256 that the catalog does not contain, with an exact real-algebraic witness.

7. **Equivariant cohomology from first principles** (Section 8). Characters from the Koszul resolution, Bott's theorem and Serre duality, with individual degrees separated by equivariant map ranks; all 27,821 published character tables are reproduced exactly. The group lift on X_{7862} has a nontrivial central commutator, giving an equivariance obstruction $\sum_i k^i \equiv 0 \pmod 2$.

8. **A downstairs search** (Section 9). From our own exhaustive upstairs survivors, 21 orbits on $X_{7862}/(\mathbb{Z}_2 \times \mathbb{Z}_2)$ admit three chiral families, no charged chiral exotics and a Higgs doublet pair. All 10 published orbits are recovered; 11 are absent from the published standard-model database.

9. **Why they were absent.** Every one of the eleven lies outside the published scan's own stated domain: its charge range, or its bound $h^1(L_a^* \otimes L_b^*) < |\Gamma|$ on vector-like pairs, which every published model satisfies.

10. **The holomorphic superpotential** (Section 10). The $U(1)^4$ selection rule with singlet insertions is exactly a network-flow membership problem (Proposition 10.1), decidable with no bound on the insertion order. It reproduces every one of the thirteen operator tables of the published database for all 2,012 models, once nine unstated conventions are identified from the data, and it shows that 954 of 2,007 published down-type tables are internally inconsistent as charge tables. Leading-order couplings are computed as genuine triple products, independent of complex structure; Γ-covariance of the trilinear form is verified exactly, and a vanishing invisible to the charges (Lemma 10.2) accounts for the only up-type rank differences with the published data.

11. **The physics verdict** (Table 9). Classifying every configuration of the 11 new and the 10 published orbits by explicit logical criteria, none is a strong candidate. Nine new orbits are ruled problematic—as are seven published ones—and two remain undecided. None of the new orbits has a leading-order top Yukawa; nine forbid dimension-five proton decay to all orders; in one, a charge-allowed R-parity-violating coupling vanishes cohomologically.

1.2 What is not claimed

The distinction between what is proved, what is necessary, and what is only observed runs through the whole paper, and we state the principal limits at the outset. No model is claimed to be a Standard Model: the downstairs spectra are $\mathcal{N} = 1$ supersymmetric and MSSM-like, the cohomology is that of a Zariski-generic member of the relevant family, and supersymmetry breaking, moduli stabilisation and physical (canonically normalised) Yukawa couplings are not addressed. Stability of the survivor sets across $K = 4, \ldots, 7$ is recorded as empirical saturation, not a finiteness theorem. Rules that are elementary or already present in the primary sources are attributed as such; our aim is a search whose conclusions can be checked, and several of its components are standard.

1.3 Organisation

Section 2 fixes the geometry and the physical conditions; Section 3 defines the logical framework. Section 4 is the search tree and every pruning rule with its proof. Section 5 gives the charge-unbounded slice results, Section 6 the cohomology certificates, Section 7 the validation. Sections 8 and 9 treat the equivariant descent and the downstairs search, Section 10 the Yukawa sector, Section 11 performance. Section 12 lists limitations and open problems. The appendices give reproduction commands and the ledger of conventions and source discrepancies.

2 Heterotic line-bundle models

2.1 Geometry

Let $A = \mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_m}$ and let $X \subset A$ be a smooth complete intersection Calabi–Yau threefold [5, 6, 7]. We assume X is *favourable*: the restrictions $J_1, \ldots, J_h$ of the hyperplane classes of the factors form an integral basis of $H^{1,1}(X)$, so $h = m$ and every line bundle is

	X_{7862} (tetraquadric)	X_{7447}
ambient space	$(\mathbb{P}^1)^4$	$(\mathbb{P}^1)^5$
defining equations	one of multidegree $(2,2,2,2)$	two of multidegree $(1,1,1,1,1)$
$(h^{1,1}, h^{2,1})$	$(4, 68)$	$(5, 45)$
d_{ijl}	2 for i,j,l distinct, else 0	2 for i,j,l distinct, else 0
$c_{2,i}$	24 for all i	24 for all i
index	$\operatorname{ind}\mathcal{O}_X(k) = 2\big(e_3(k) + e_1(k)\big)$	$\operatorname{ind}\mathcal{O}_X(k) = 2\big(e_3(k) + e_1(k)\big)$
$d_X = \gcd \operatorname{ind}$	2	2
anomaly constant C_X (Eq. (6))	48	40
free actions used	$\mathbb{Z}_2$, $\mathbb{Z}_2 \times \mathbb{Z}_2$	$\mathbb{Z}_2$

Table 1: The two benchmark geometries. e_j denotes the j-th elementary symmetric polynomial in the components of k. Topology was recomputed from the configuration matrices and agrees with the published values for all 68 favourable CICYs (c_2: 68/68, Euler number: 68/68).

$\mathcal{O}_X(k) \coloneqq \mathcal{O}_A(k)|_X$ for a unique $k \in \mathbb{Z}^h$. The topological data are the triple intersection numbers and the second Chern class,

$$d_{ijl} = \int_X J_i J_j J_l \in \mathbb{Z}, \qquad c_{2,i} = \int_X c_2(TX)\, J_i, \tag{1}$$

both computed exactly from the ambient Chow ring and the tangent–normal sequence. We write

$$D(x,y,z) = d_{ijl} x^i y^j z^l, \qquad q_r(k) = d_{rij} k^i k^j, \qquad M(k)_{jl} = d_{ijl} k^i, \tag{2}$$

so that $M : \mathbb{Z}^h \to \mathrm{Sym}_h(\mathbb{Z})$ is a group homomorphism. By Hirzebruch–Riemann–Roch the index of a line bundle is

$$\operatorname{ind}\mathcal{O}_X(k) = \sum_q (-1)^q h^q(X, \mathcal{O}_X(k)) = \tfrac{1}{6} D(k,k,k) + \tfrac{1}{12} c_{2,i} k^i. \tag{3}$$

2.2 Line-bundle sums and the upstairs conditions

Let $V = \bigoplus_{a=1}^5 L_a$, $L_a = \mathcal{O}_X(k_a)$. With V embedded in one E_8 factor, its structure group $S(U(1)^5) \subset SU(5)$ leaves an $SU(5)$ GUT times $U(1)^4$ in four dimensions. The upstairs conditions of [3] on the charge matrix $K = (k_1, \ldots, k_5)$ are:

(C1) $c_1(V) = \sum_a k_a = 0$.

(C2) No nonempty proper subset $S \subsetneq \{1, \ldots, 5\}$ has $\sum_{a \in S} k_a = 0$, so the structure group is not smaller than $S(U(1)^5)$.

(C3) $\operatorname{ind} V = \operatorname{ind} \wedge^2 V = -3|\Gamma|$, giving three chiral families after the quotient by a freely acting Γ.

(C4) $c_2(TX) - c_2(V)$ is effective (trivial hidden bundle). Since the J_r are nef, a necessary condition is $c_{2,r} - c_2(V)_r \geq 0$ for all r. With (C1), $c_2(V)_r = -\frac{1}{2} \sum_a q_r(k_a)$.

(C5) V is poly-stable: there is a Kähler class $J = t^i J_i$ at which all five slopes $\mu_{k_a}(t) = t^\top M(k_a) t$ vanish simultaneously.

(C6) Bogomolov: $c_2(V) \cdot J \geq 0$ at that class.

(C7) For the target `GUT_FULL`: $h^\bullet(X, V) = (0, 3|\Gamma|, 0, 0)$ and $h^\bullet(X, \wedge^2 V) = (0, 3|\Gamma| + n_h, n_h, 0)$ with $n_h > 0$ (no anti-families; at least one Higgs pair).

Throughout, the Kähler parameter is restricted to the positive ambient chamber $T_+ = \{t \in \mathbb{R}^h : t^i > 0\}$; for both benchmarks this is the declared Kähler cone, an input assumption carried in every certificate.

2.3 Descent and the downstairs spectrum

Let Γ be a freely acting abelian group, $\hat{X} = X/\Gamma$. An equivariant structure on each L_a lets V descend; a Wilson line, i.e. a homomorphism $\Gamma \to U(1)_Y \subset SU(5)$, breaks $SU(5) \to SU(3)\times SU(2)\times U(1)_Y$. Upstairs, $H^1(X,V)$ carries the **10**s, $H^1(X,\wedge^2 V)$ the $\overline{\mathbf{5}}$s, $H^2(X,\wedge^2 V) \cong H^1(X,\wedge^2 V^*)^*$ the **5**s, and $H^1(X, L_a \otimes L_b^*)$ the singlets S_{ab}. Downstairs fields are the Γ-invariant parts of these cohomologies tensored with the Wilson-line representation. The target DOWNSTAIRS_MSSM requires three chiral families of Q, u^c, e^c, d^c, L, no charged chiral exotics, and at least one Higgs doublet pair; the distinct stricter target of an exact non-supersymmetric Standard Model is not addressed.

2.4 The search problem

For a bound $K \in \mathbb{Z}_{>0}$ let

$$\mathcal{D}_K = \{K \in \mathbb{Z}^{h\times 5} \ : \ |k_a^i| \le K,\ \textstyle\sum_a k_a = 0\}/S_5, \tag{4}$$

the charge matrices with entries in the box, up to permutation of the summands. The *bounded search problem* is to determine the subset of $\mathcal{D}_K$ satisfying (C1)–(C7). Its nominal size is enormous: already at $K = 7$ on X_{7862} there are $15^{16} \approx 6.6 \times 10^{18}$ ordered choices of the first four columns. The *unbounded problem*—to decide the full solution set in $\mathbb{Z}^{h\times 5}$—is the real question, and Section 5 is our partial answer to it.

3 Logical framework: what a search result means

Every condition and every output of the engine carries one of six logical statuses, and they are never mixed.

Definition 3.1 (Statuses of conditions)**.** A condition used by the search is

- EXACT if it is equivalent to the property it tests;
- NECESSARY if every valid model satisfies it (so failing it proves invalidity, but passing it proves nothing);
- SUFFICIENT if satisfying it proves the property (e.g. an exact rational or real-algebraic slope witness);
- NUMERIC_WITNESS if it is supported only by floating-point evidence;
- HEURISTIC if it has no proof in either direction;
- UNKNOWN if it could not be decided.

Only EXACT and NECESSARY conditions may prune. Only EXACT and SUFFICIENT conditions may certify acceptance. HEURISTIC and NUMERIC_WITNESS conditions may do neither.

Definition 3.2 (Search outcomes)**.** For a declared domain $\mathcal{D}$ (geometry, $|\Gamma|$, target, Kähler chamber, and either a box K or a slice $p^\perp$):

- UNSAT_BOUNDED: every state of the finite domain was accounted for and none satisfies the declared necessary conditions;
- UNKNOWN_BOUNDED: the domain is exhausted and necessary-condition survivors remain, none certified;

- SAT_BOUNDED: a survivor is certified by sufficient conditions (never emitted: no model reaches that standard here);
- UNSAT_SLICE_UNBOUNDED: an entire slice $p^{\perp}$ has been exhausted over *all* integer charges, with no survivor (Section 5);
- UNSAT_UNBOUNDED: no model exists for any charges (only the divisibility obstruction, Proposition 4.12, emits this).

Stability of a survivor set as K grows is recorded as EMPIRICAL_SATURATION and never converted into an unbounded statement.

The following observation is elementary but it is what makes the whole design safe, and it answers precisely when an incomplete test may be used.

Lemma 3.3 (Soundness of pruning by necessary conditions)**.** *Let $\mathcal{D}$ be finite and let $R_1, \dots, R_n$ be conditions each of which is necessary for the declared target under stated assumptions. If the search enumerates $\mathcal{D}$ and discards a state only when some R_i fails—on the state or on a prefix from which the failure provably propagates to every completion—then every valid model in $\mathcal{D}$ survives. In particular UNSAT_BOUNDED is correct. Replacing any R_i by a weaker necessary condition can only enlarge the survivor set.*

Proof. A valid model satisfies every R_i, so it is never discarded at the leaf. A prefix is discarded only if every completion fails some R_i, which a valid model does not. Weakening a necessary condition removes rejections and adds none. □

The last sentence is the answer to a question that matters in practice: an *incomplete* test (one that misses some obstructions) is harmless for UNSAT_BOUNDED; only an *unsound* test (one that rejects a state that could be valid) is dangerous. The finite coefficient box of the slope test (Section 4.3) is incomplete but sound, and may therefore contribute to UNSAT_BOUNDED; its completion is nevertheless worth having, because an incomplete test leaves false survivors, and Section 7 shows that it does.

4 The search tree and how it is pruned

4.1 Canonical enumeration

Summand permutations act on $\mathcal{D}_K$, so we fix a total order on $\mathbb{Z}^h$ (lexicographic) and represent each class by its sorted column tuple $k_1 \preceq k_2 \preceq \cdots \preceq k_5$, repetitions allowed. Because $\sum_a k_a = 0$, the first four columns determine the fifth. The search is a depth-first traversal of the tree whose depth-m nodes are sorted prefixes $(k_1, \dots, k_m)$, $m \le 4$; the children of a node append a column $k_{m+1} \succeq k_m$ from a precomputed pool, and a depth-4 node produces the single leaf $k_5 = -\sum_{a \le 4} k_a$, kept only if it lies in the box and satisfies $k_5 \succeq k_4$. Every class of $\mathcal{D}_K$ is reached exactly once, which gives a direct completeness argument; ambient-factor permutations are *not* quotiented during the search but only afterwards, when orbits are compared (Section 7).

4.2 The rules

Table 2 lists every rule, its status and where it acts. We give the proofs of those that are not immediate.

R2 (single-index range). At a common zero of the slopes, a nontrivial line bundle L_a has $h^0 = h^3 = 0$, since a line bundle of slope zero with a nonzero section is trivial, and (C2) excludes trivial summands. No anti-families means $h^2(X, V) = 0$, hence $h^2(L_a) = 0$ and $\operatorname{ind} L_a = -h^1(L_a) \le 0$. The five indices sum to $-3|\Gamma|$, so each lies in $[-3|\Gamma|, 0]$, and so does every partial sum (R5).

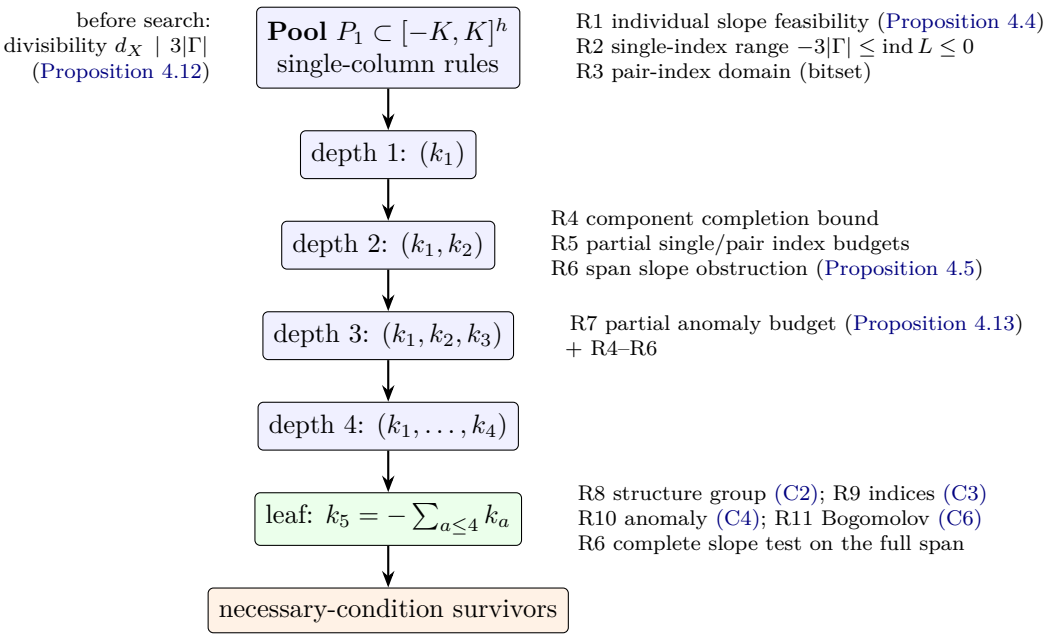


Figure 1: The search tree. Each rule is a necessary condition (Lemma 3.3); rules R4, R5 and R6 are monotone and prune entire subtrees. Rules R3 and R4 are enforced by construction through bitset-indexed candidate lists rather than tested inside the loop.

R4 (component completion). With m columns fixed, the remaining $5 - m$ columns have entries in $[-K, K]$ and must cancel the partial sum, so $|\sum_{a \le m} k_a^i| \le (5 - m)K$ by the triangle inequality.

4.3 The slope obstruction

Poly-stability of a sum of line bundles with $c_1(V) = 0$ means a common zero of the five slopes:

$$\exists\, t \in T_+ : \qquad \mu_{k_a}(t) = t^\top M(k_a)\, t = 0 \qquad (a = 1, \ldots, 5). \tag{SZ}$$

Lemma 4.1 (Sign obstruction)**.** *If $M \in \mathrm{Sym}_h(\mathbb{Z})$ is nonzero and entrywise nonnegative (or nonpositive), then $t^\top M t \neq 0$ for every $t \in T_+$.*

Proof. Each term $M_{jl} t^j t^l$ has the same weak sign and at least one is strictly nonzero, because $t^j t^l > 0$. □

Proposition 4.2 (Combined test)**.** *If (SZ) holds then for every $c \in \mathbb{R}^5$ the matrix $M(\sum_a c_a k_a) = \sum_a c_a M(k_a)$ is zero or has entries of both signs. Contrapositively, a single $c \in \mathbb{Z}^5$ for which $M(\sum_a c_a k_a)$ is nonzero and one-signed proves that V is not poly-stable anywhere in T_+.*

Proof. At the witness t_*, $t_*^\top M(\sum_a c_a k_a) t_* = \sum_a c_a \mu_{k_a}(t_*) = 0$; apply Lemma 4.1. □

The scan of [3] applies Proposition 4.2 with $c \in \{-1, 0, 1\}^5$. This is *sound* but not *complete*: passing the test for all c in a finite box does not imply (SZ). (Since $\sum_a k_a = 0$, $M(\sum_a c_a k_a)$ depends only on the class of c modulo $\mathbb{Z}(1, \ldots, 1)$; the 121 vectors of the box with first nonzero entry positive fall into 106 classes, and the box is exactly the set of classes of spread $\max c - \min c \le 2$.) The quantified statement, however, is decidable exactly.

Proposition 4.3 (Exact obstruction cone)**.** *Let $S = \mathrm{span}_{\mathbb{R}}\{k_1, \ldots, k_5\} \subseteq \mathbb{R}^h$ and $P = \{k \in \mathbb{R}^h : M(k)_{jl} \ge 0\ \forall j, l\}$, a rational polyhedral cone. Some $c \in \mathbb{Z}^5$ yields a one-signed nonzero combined matrix if and only if $S \cap P$ contains a point k with $M(k) \neq 0$.*

	Rule	Status	Acts at
R1	μ_k has a zero in T_+ (signed-matrix test; exact per Proposition 4.4)	exact (bench.)	pool
R2	$-3\lvert\Gamma\rvert \le \operatorname{ind}\mathcal{O}_X(k) \le 0$ (GUT_FULL)	necessary	pool
R3	declared domain: $\operatorname{ind}(L_a \otimes L_b) \le 0$ for all pairs	domain restriction	pool, prefix
R4	$\lvert\sum_{a\le m} k_a^i\rvert \le (5-m)K$	necessary	prefix
R5	partial sums of single and pair indices lie in $[-3\lvert\Gamma\rvert, 0]$	necessary	prefix
R6	$\operatorname{span}_{\mathbb{R}}\{k_a\} \cap P = \{0\}$ (complete slope obstruction)	necessary	prefix, leaf
R7	partial anomaly budget	necessary	prefix
R8	no proper zero-sum subset (C2)	exact	leaf
R9	$\operatorname{ind} V = -3\lvert\Gamma\rvert$ ($\operatorname{ind}\wedge^2 V$ then automatic, Proposition 4.11)	exact	leaf
R10	$c_{2,r} - c_2(V)_r \ge 0$	necessary	leaf
R11	Bogomolov pairing	necessary	leaf
R12	$d_X \mid 3\lvert\Gamma\rvert$	exact, unbounded	before search

Table 2: Pruning rules. R3 restricts the declared domain to the historical domain of [3]; it is not claimed necessary, and the independent replays (Section 7) also run with it removed. R1 is exact on geometries with $d_{ijj} = 0$, which includes both benchmarks.

Proof. M is linear, so $\{M(\sum_a c_a k_a) : c \in \mathbb{R}^5\} = M(S)$, and P is the preimage of the entrywise-nonnegative matrices; $-c$ handles the nonpositive case. A rational point of the rational cone $S \cap P$ can be scaled to an integer point k, and c is recovered by solving the consistent rational system $\sum_a c_a k_a = k$, then clearing denominators. □

On geometries with $d_{ijj} = 0$ for all i, j—both benchmarks—$M(k)$ has zero diagonal, and nothing is lost per coefficient vector:

Proposition 4.4 (Per-coefficient optimality)**.** *If $d_{ijj} = 0$ for all i, j, then for any $k \in \mathbb{Z}^h$, the form μ_k has a zero in T_+ if and only if $M(k)$ is zero or has entries of both signs.*

Proof. ($\Rightarrow$) is Lemma 4.1. ($\Leftarrow$): the diagonal vanishes, so suppose $M_{jl} > 0$ and $M_{j'l'} < 0$ at off-diagonal positions. Let $t(T)$ equal T at positions j, l and 1 elsewhere; then $\mu_k(t(T)) = 2M_{jl}T^2 + O(T) \to +\infty$. The analogous path at (j', l') gives $-\infty$. T_+ is connected and μ_k continuous, so μ_k vanishes in T_+. □

In particular the single-bundle condition "$\mu(L) = 0$ somewhere in the interior of the Kähler cone" is decided exactly on these geometries by the signed-matrix test (R1), and the only incompleteness of the finite test is the truncation of the c-search, which Proposition 4.3 removes.

Proposition 4.5 (Monotonicity)**.** *If $S' \subseteq S$ and $S' \cap P$ contains a point with $M \neq 0$, so does $S \cap P$. Hence the test applied to the span of any prefix $(k_1, \ldots, k_m)$ prunes every completion of that prefix.*

Proof. Immediate from $S' \cap P \subseteq S \cap P$. □

To decide $S \cap P \neq \{0\}$ exactly we need P to be pointed and its extreme rays.

Lemma 4.6 (The tetraquadric cone)**.** *On X_{7862}, $M(k)_{jl} = 2(k^m + k^n)$ for $j \neq l$, where $\{m, n\}$ is the complement of $\{j, l\}$ in $\{1, 2, 3, 4\}$, and $M(k)_{jj} = 0$. Hence $P = \{k \in \mathbb{R}^4 : k^m + k^n \ge 0\ \forall m < n\}$, $M(k) = 0$ only for $k = 0$, and P is pointed with exactly eight extreme rays: the coordinate vectors e_i and the four vectors with one entry -1 and the others $+1$.*

Proof. The formula follows from $d_{ijl} = 2$ for distinct indices. If all pairwise sums vanish then $k^1 + k^2 = k^1 + k^3 = k^2 + k^3 = 0$ forces $k^1 = k^2 = k^3 = 0$ and then $k^4 = 0$, so $P \cap (-P) = \{0\}$. A ray of P is extreme iff the constraints tight on it have rank three. For e_1 the tight constraints are the three pairs not containing 1; for $(-1, 1, 1, 1)$ the three pairs containing 1; each set has rank three. Enumerating the $\binom{6}{3} = 20$ triples of constraints shows there are no others. □

For X_{7447} the analogous computation (exact, by enumeration) gives $2h = 10$ extreme rays: the e_i and the vectors with one entry $-2 = -(h-3)$ and the others $+1$. In general the implementation enumerates extreme rays of $S \cap P$ exactly in integer arithmetic and declines to prune if the pointedness check fails.

The primal enumeration costs a number of kernel computations that grows with the dimension of S and dominated the running time on X_{7447} (90% of a profiled scan). The dual form is much cheaper exactly where the search spends its time.

Proposition 4.7 (Dual decision)**.** *Let P be pointed with extreme rays $r_1, \dots, r_s$, let A be a matrix whose rows span $S^\perp$, and $d = \dim S^\perp = h - \operatorname{rk} S$. Then*

$$S \cap P = \{0\} \iff \exists\, \varphi \in S^\perp : \ \langle \varphi, r_i \rangle > 0 \ \forall i \iff 0 \notin \operatorname{conv}\{Ar_1, \dots, Ar_s\} \subset \mathbb{R}^d.$$

Proof. As P is pointed its base $B = \operatorname{conv}\{r_i\}$ is compact with $0 \notin B$. If $S \cap P = \{0\}$ then $S \cap B = \emptyset$, and the compact convex set B is strictly separated from the closed subspace S by a functional that must vanish on S, i.e. lies in $S^\perp$, and is positive on B. Conversely such a φ vanishes on S and is positive on $P \setminus \{0\}$. The last equivalence is Gordan's alternative [8] for the projected points Ar_i. □

The dimension d *shrinks* as the prefix grows. At the deepest and most numerous nodes $d \in \{1, 2\}$: for $d = 1$ the test is whether the projected scalars take both signs ($O(s)$), for $d = 2$ an angular-gap test ($O(s \log s)$). The primal enumeration remains the reference implementation, the fallback for $d > 2$, and the producer of the explicit integer witness k_c recorded in every rejection certificate. The two were differentially tested on 12,000 random column sets across every span rank with no disagreement.

Example 4.8 (A spurious survivor of the finite test)**.** On X_{7862} with $\Gamma = \mathbb{Z}_2$ at $K = 3$, the charge matrix with columns

$$(-3, -1, 1, 2),\ (0, 1, -1, 0),\ (0, 1, -1, 0),\ (1, 1, -1, -1),\ (2, -2, 2, -1)$$

satisfies every other declared condition and passes the test for all $c \in \{-1, 0, 1\}^5$; a numerical optimiser stalls at slope residual 3.6×10^{-7}. The complete test rejects it with the integer witness $k_c = (1, 0, 0, 0) \in S$, for which $M(k_c)$ is entrywise nonnegative and nonzero. It is absent from the published catalog. Its orbit is the only difference between the finite and the complete test at $K = 3$ (8 versus 7 orbits) and at $K = 4$ (11 versus 10), and removing it is what gives exact catalog agreement (Section 7).

Soundness was tested by a counterexample search: 12,000 randomly generated bundles carrying an exact common slope zero on X_{7862} and X_{7447} were never rejected; an independently written Fourier–Motzkin implementation agreed with the extreme-ray implementation on 3,000 random column sets; and every published model and every survivor with an exact slope witness is retained.

4.4 Beyond sign tests: a Pólya hierarchy and an exact decision

The sign test is exact per combination only when $d_{ijj} = 0$ (Proposition 4.4), and even then a bundle can pass every sign test while its five slopes have no *simultaneous* zero. Two further exact tools address this; both produce certificates checked in exact arithmetic.

Proposition 4.9 (Pólya certificates)**.** *If for some c and some $N \geq 0$ the form $(t_1 + \cdots + t_h)^N \sum_a c_a \mu_{k_a}(t)$ has all coefficients nonnegative and not all zero, then (SZ) has no solution. Every form strictly positive on the closed simplex admits such an N; $N = 0$ is the sign test.*

Proof. A nonzero polynomial with nonnegative coefficients is positive on the open orthant, and $(\sum_i t_i)^N > 0$ there; at a common zero the combination would vanish. The converse direction is Pólya's theorem. □

The coefficients are linear in $k_c = \sum_a c_a k_a$, so each level of the hierarchy is again a polyhedral cone and a certificate is found by linear programming and verified exactly. On geometries with $d_{ijj} = 0$ the hierarchy provably adds nothing beyond $N = 0$ (by Proposition 4.4); on the other catalog geometries we tested it added no certificate for charges in $[-2, 2]$ up to $N = 4$. The obstruction for the bundles below, where it exists, is therefore *joint*.

Proposition 4.10 (Exact decision in rank $h - 1$)**.** *Suppose* $\operatorname{rk} K = h - 1$. *Set* $t_h = 1$, *which loses no interior point, and compute a lexicographic Gröbner basis of* $h - 1$ *independent slopes. If it is in shape position,* $t_i = g_i(x)$ *and* $p(x) = 0$*; isolate every real root of every irreducible factor of* p *in an exact rational interval and enclose the other coordinates by rigorous interval arithmetic. A root at which all coordinates are positive, and modulo whose minimal polynomial every slope vanishes identically, is an exact real-algebraic witness of poly-stability. If no real root is interior,* (SZ) *has no solution.*

Proof. In rank $h - 1$ the common zeros with $t_h = 1$ are the solutions of the $h - 1$ independent equations, which in shape position are finitely many and given exactly by the roots of p; interval enclosures that exclude zero decide each coordinate's sign rigorously. □

When the basis is not in shape position the decision proceeds factor by factor: factors of p without a positive real root are discarded (an interior point has $t_{h-1} > 0$); a rational root is substituted and the smaller system decided recursively; over an irrational factor f the remaining coordinates are solved linearly in $\mathbb{Q}[x]/(f)$ when possible. Anything else, including positive-dimensional zero sets (rank below $h - 1$), is reported as UNDECIDED. On the tetraquadric the procedure certifies the published model with witness $t = (1, 1, 1, 1)$, and the spurious candidate of Example 4.8, whose zero set is positive-dimensional, is excluded by the sign test instead: the tools are complementary.

4.5 Index identity and divisibility

Proposition 4.11 (Index identity)**.** *On any Calabi–Yau threefold, if* $V = \bigoplus_{a=1}^{5} L_a$ *has* $c_1(V) = 0$ *then* $\operatorname{ind} \wedge^2 V = \operatorname{ind} V$.

Proof. The linear terms vanish: $\sum_a c_2 \cdot k_a = 0$ and each a occurs in four pairs. For the cubic terms, $\sum_{a<b} D(k_a + k_b)^3 = 4 \sum_a D(k_a)^3 + 3 \sum_{a \neq b} D(k_a, k_a, k_b)$ and $\sum_{b \neq a} k_b = -k_a$, so the sum is $4 \sum_a D(k_a)^3 - 3 \sum_a D(k_a)^3$. □

This is stated in [3] after their equation (4.7); we use it to drop a redundant filter and keep it as a free consistency check. (For rank n the coefficient is $n - 4$.)

Proposition 4.12 (Divisibility obstruction)**.** *Let* $d_X = \gcd\{\operatorname{ind} \mathcal{O}_X(k) : k \in \mathbb{Z}^h\}$. *If* $d_X \nmid 3|\Gamma|$, *no sum of line bundles on* X *has* $\operatorname{ind} V = -3|\Gamma|$, *for any charges. Moreover* $d_X = \gcd\{\operatorname{ind} \mathcal{O}_X(k) : k \in \{0, 1, 2, 3\}^h\}$.

Proof. $\operatorname{ind} V = \sum_a \operatorname{ind} L_a \in d_X \mathbb{Z}$. For the second claim, ind is an integer-valued polynomial of degree three, hence a $\mathbb{Z}$-combination of $\prod_i \binom{k_i}{\alpha_i}$ with $|\alpha| \leq 3$; its values generate the same subgroup as the coefficients, which are integer combinations of finite differences at points of $\{0, 1, 2, 3\}^h$. □

Over the 68 favourable CICYs, $d_X \in \{1, 2, 3, 5\}$ occurs $19, 44, 4, 1$ times. Combined with the symmetry orders recorded for each geometry, the obstruction predicts *zero* models for exactly two pairs, $(7761, |\Gamma| = 2)$ with $d_X = 5$ and $(7447, |\Gamma| = 5)$ with $d_X = 2$; the published catalog

has zero models for both, and no catalog entry with models contradicts it. The prediction is not vacuous: geometries with odd $|\Gamma|$ and $d_X \mid 3|\Gamma|$ do carry models. The rule is elementary; we have not found it stated in the sources we examined. It is the only rule here whose conclusion carries no bound on the charges.

4.6 The partial anomaly budget

Proposition 4.13. *Assume (C1), a trivial hidden bundle and nef J_r. Then (C4) implies the separable inequality*

$$\sum_{a=1}^{5} q_r(k_a) \;\geq\; -2c_{2,r} \qquad \text{for all } r. \tag{5}$$

Let P_1 be a pool containing every column any valid completion can use, and $Q_r = \max_{k\in P_1} q_r(k)$. If for a prefix $(k_1, \ldots, k_m)$ and some r

$$\sum_{a\leq m} q_r(k_a) + (5-m)Q_r < -2c_{2,r},$$

no completion satisfies (5) and the branch is pruned.

Proof. With $\sum_a k_a = 0$ the Whitney formula gives $c_2(V)_r = -\frac{1}{2}\sum_a q_r(k_a)$, so $c_{2,r} - c_2(V)_r \geq 0$ is (5). Each remaining column contributes at most Q_r. □

The pool in the implementation is the retained single-column list, itself defined by necessary conditions, and the derived fifth column is required to lie in it. The bound is inactive for $K \leq 2$, since small charges cannot reach it, and first bites at $K = 3$ (Table 4).

4.7 Implementation

Two features matter for the complexity. First, the retained pool is indexed by bitsets: for each component i and threshold, the set of columns whose component lies above or below it, and for each column the set of compatible partners under R3. Rules R3 and R4 are then enforced *by construction*—a node iterates only over the intersection of the relevant bitsets—rather than tested inside the loop. This cut fourth-column attempts by factors of 19–28 and wall time by 3–7.5× at identical survivor sets, which was verified by re-running every recorded domain and requiring exact set equality. Second, the span of the prefix is maintained incrementally in a fraction-free integer row-reduced form, which is also the memoisation key for the slope decision, so repeated spans are decided once. Every run is checkpointed; a checkpoint embeds the SHA-256 of the source and inputs and refuses to resume under changed code.

4.8 Measured effect of the pruning

Table 3 shows the funnel at $K = 2$ and Table 4 the effect of each rule and of K.

Two observations. The complete slope rule, applied at every depth, removes 2,730 of the 10,573 depth-2 prefixes already at $K = 2$, and cuts the bundles built by factors of 2.8 at $K = 3$ and 2.5 at $K = 4$. And the pruning improves with K rather than degrading: from $K = 1$ to 3 the nominal domain grows by factors $3,544$ and 218 while the bundles built grow by 30 and 5.2.

5 Searches with no charge cutoff

A box search says nothing about charges outside the box. This section shows that for an explicit and large part of the landscape the box can be removed altogether.

stage (X_{7862}, $\Gamma = \mathbb{Z}_2$, $K = 2$, finite slope test, pre-indexing engine)	count
ordered four-column assignments (nominal)	152,587,890,625
individual columns in the box	625
retained single columns (R1, R2)	235
retained depth-2 prefixes	10,573
retained depth-3 prefixes	123,629
fourth-column attempts	7,875,639
retained depth-4 prefixes	252,154
complete canonical bundles	38,161
no proper zero-sum subset (R8)	24,043
ind $V = -6$ (R9)	6,261
all pair indices in range (R3)	1,356
anomaly (R10)	1,260
slope (R6)	**120**

Table 3: Funnel of the search. Prefix counters count prefixes, not full states, and must not be added as if they partitioned the nominal domain. The ratio of nominal assignments to materialised bundles, about 4×10^6, includes canonicalisation and is not a pure physics pruning factor; at the last level the slope obstruction alone removes a factor 10.5.

K	slope rule	wall (s)	RSS (MiB)	4th-col. attempts	bundles built	survivors	orbits
1	finite	0.05	25.4	4,636	713	0	0
1	complete	0.09	25.3	2,535	334	0	0
2	finite	4.00	33.2	420,266	38,161	120	5
2	complete	3.99	37.2	170,121	10,097	120	5
3	finite	15.88	50.5	2,304,135	146,425	192	**8**
3	complete	18.42	68.3	963,549	52,800	168	7
4	finite	36.65	65.6	7,147,830	294,130	264	**11**
4	complete	43.62	105.6	3,236,246	118,577	240	10
5	complete	85.44	141.0	6,537,468	162,761	240	10
6	complete	118.88	193.8	10,199,673	206,847	240	10
7	complete	179.15	188.9	14,516,351	231,946	240	10

Table 4: X_{7862}, $\Gamma = \mathbb{Z}_2$, target `GUT_FULL`, bitset-indexed engine. Orbits are ambient-row orbits of the column-canonical survivors. The finite test retains one spurious orbit at $K = 3$ and $K = 4$ (Example 4.8). With the dual decision of Proposition 4.7 the complete rule at $K = 4$ takes 28.8 s, faster than the finite test. The partial anomaly budget (Proposition 4.13) first acts at $K = 3$, pruning 3,734 depth-3 and 166,040 depth-4 branches under the complete rule and reducing built bundles from 75,534 to 52,800.

5.1 The slice decomposition

Write $q_i(t) = d_{ijl} t^j t^l$, so that $\mu_k(t) = \langle k, q(t) \rangle$.

Proposition 5.1 (Slices)**.** *Suppose $q_i(t) > 0$ on T_+ for every i (on both benchmarks $q_i(t) = 4e_2(t_{\neq i})$). If $V = \bigoplus_a L_a$ satisfies (SZ), there is a primitive $p \in \mathbb{Z}^h$ with every $p_i > 0$ such that $\langle k_a, p \rangle = 0$ for all a.*

Proof. At a witness t, $u = q(t) \in \mathbb{R}^h_{>0}$ and $\langle k_a, u \rangle = \mu_{k_a}(t) = 0$, so $u \in S^\perp$ with $S = \operatorname{span}_{\mathbb{R}}\{k_a\}$. S is spanned by integer vectors, so $S^\perp$ is a rational subspace; it meets the open positive orthant (at u), and rational points are dense in it, so it contains a positive rational vector. Clear denominators and divide by the gcd. □

Corollary 5.2. $\operatorname{rk} K \leq h - 1$ *for every slope-feasible charge matrix. (Confirmed on data: all 240 X_{7862} survivors at $K \leq 7$ have rank 3 or 2, never 4.)*

The decomposition is exhaustive: every slope-feasible model lies in the slice $p^\perp$ of at least

one positive primitive p. One search over $\mathbb{Z}^{h\times 5}$ becomes a family of searches over rank-$(h-1)$ lattices.

5.2 A charge bound on good slices

Summing (5) over r gives, with $s(k)=\sum_i k^i$,

$$\sum_{a=1}^{5} F(k_a) \le C_X, \qquad F(k)=|k|^2-s(k)^2, \qquad C_X=\frac{\sum_r c_{2,r}}{h-2}, \tag{6}$$

because on both benchmarks $\sum_r q_r(k)=2(h-2)(s(k)^2-|k|^2)$; hence $C_X=48$ for X_{7862} and 40 for X_{7447}.

Proposition 5.3 (Per-slice bound)**.** *Let p be a slice direction with*

$$\Big(\sum_i p_i\Big)^2 > (h-1)\,|p|^2. \tag{7}$$

Then F is positive definite on $p^\perp$, and every column of every model in the slice satisfies

$$|k_a|^2 \;\le\; R^2(p) := \frac{C_X\,|p|^2}{(\sum_i p_i)^2-(h-1)|p|^2}.$$

Proof. For $k\in p^\perp$ let w be the orthogonal projection of $\mathbf{1}=(1,\dots,1)$ onto $p^\perp$. Then $s(k)=\langle k,w\rangle$, so $s(k)^2\le |k|^2|w|^2$ and $F(k)\ge |k|^2(1-|w|^2)$. Since $|w|^2=h-(\sum p)^2/|p|^2$, condition (7) is exactly $|w|<1$. Every summand in (6) is then nonnegative, hence bounded by C_X, and $|k_a|^2(1-|w|^2)\le C_X$. □

Corollary 5.4 (A search with no charge cutoff)**.** *For p satisfying (7) the admissible column pool $P(p)=\{k\in\mathbb{Z}^h : \langle k,p\rangle=0,\ F(k)\le C_X\}$ is finite and exactly computable, by walking the ball $|k|^2\le R^2(p)$ and solving the linear condition for the last coordinate. Searching $P(p)$ with the rules of Section 4 exhausts the entire slice over all integer charges. The outcome* UNSAT_SLICE_UNBOUNDED *carries no charge bound.*

5.3 Results on the tetraquadric

On X_{7862} with $\Gamma=\mathbb{Z}_2$ and target GUT_FULL, of the primitive positive p with entries ≤ 10 (up to permutations of the ambient factors), 469 satisfy (7). All were exhausted in 36.7 s in total:

slices satisfying (7), entries ≤ 10	469
proven empty for all charges (UNSAT_SLICE_UNBOUNDED)	**425**
containing models	44
ambient-row orbits found, maximum charge	6 orbits, max $\lvert k\rvert = 3$
largest radius $R^2(p)$	2,484
column pool sizes	85–700

Pools stay small although radii grow, because the lattice $p^\perp$ thins as $|p|$ grows.

5.4 Where the difficulty is

The slices that fail (7) are not an artefact of the method: real models live there. For $p=(1,1,3,4)$, $(\sum p)^2=3|p|^2$ exactly, and $p=(1,1,4,4)$ has $100<102=3|p|^2$. There F is degenerate or indefinite on $p^\perp$, the pool is genuinely infinite, and the anomaly bounds nothing. What stops the charges on these slices is the *index*, which grows cubically along the directions with $F\le 0$: on the null ray of $p=(1,1,3,4)$, ind takes the values $2, 52, 198$ at

$k = (-2,-2,0,1), (-4,-4,0,2), (-6,-6,0,3)$, leaving the window $[-3|\Gamma|, 0]$ at once. More generally $F(k) \leq 0$ forces $s(k) \neq 0$ (if $s(k) = 0$ then $F(k) = |k|^2 > 0$), so on X_{7862} $\mathrm{ind}(nk) = 2(n^3 e_3(k) + n\, s(k))$ is unbounded along every such ray. Converting this ray-wise statement into a bound on the semialgebraic set $\{F \leq 0,\ \mathrm{ind} \in [-3|\Gamma|, 0]\}$ is the open step; we state it as a problem in Section 12.

Probing the 22 uncovered slices with entries ≤ 6 at an in-slice box $K = 16$, far beyond the full-box reach, over pools with thousands of large-charge columns, produces maximum charge 4 and nothing new. The 84 proven slices with entries ≤ 6 together with these 22 probed ones reproduce *exactly* the 10 orbits of the full $K = 7$ search and of the published catalog.

5.5 A conditional bound

Conjecture 5.5 (Charge bound from anomaly and slope)**.** *Writing $u = q(t)$ for a slope witness and $\delta(u)^2 = h - (\sum_i u_i)^2/|u|^2$, the argument of Proposition 5.3 gives $\sum_a |k_a|^2(1-\delta(u)^2) \leq C_X$, a bound whenever $\delta(u) < 1$. On X_{7862} the infimum of $(\sum u)^2/|u|^2$ over the attainable u is exactly 3, approached as one $t_i \to 0$, so $\delta \to 1$ and the bound degenerates near the boundary of the Kähler cone. We conjecture that the index condition keeps slope witnesses of index-admissible models uniformly away from that boundary, which would make the set of models on X_{7862} finite by this route.*

We make no finiteness claim. Finiteness of the *topology* of V is argued in [3]; the conjecture concerns the charges themselves.

6 Certified line-bundle cohomology

The spectrum condition (C7) needs $h^q(X, L)$ for the line bundles L_a and $L_a \otimes L_b$. We compute these from the Koszul resolution of $\mathcal{O}_X$ in A [7, 9],

$$E_1^{-p,q} = H^q(A,\ L \otimes \wedge^p N^*) \implies H^{q-p}(X, L),$$

where each E_1 term is the cohomology of a line bundle on a product of projective spaces, given by Bott's formula [10] and Künneth. (Closed formulae are known for several CICYs [11], and the map ranks of these Koszul complexes have been studied through Hilbert functions [12]; we use neither as input, only the exact spectral sequence.)

Collapse. The differential d_r has bidegree $(r, 1-r)$. If no differential can join two occupied E_1 nodes, the sequence degenerates at E_1 and each $h^q(X, L)$ is read off. This bidegree-support test is exact.

One differential. When exactly two nodes are occupied and joined by one differential Φ, the cohomology is $\ker \Phi$ and $\mathrm{coker}\, \Phi$. Φ is induced by the defining equations, so in the fixed monomial bases of the ambient cohomology it is a matrix $\Phi(c)$ whose entries are $\mathbb{Z}$-linear in the coefficient vector c of the equations.

Proposition 6.1 (Generic rank from one finite-field minor)**.** *Let $\Phi(c) \in \mathrm{Mat}_{m\times n}(\mathbb{Z}[c])$ with entries polynomial in c, let $c_0 \in \mathbb{Z}^N$, p a prime, and suppose $\mathrm{rk}_{\mathbb{F}_p}(\Phi(c_0) \bmod p) \geq r$. Then (i) some $r \times r$ minor $\delta \in \mathbb{Z}[c]$ has $\delta(c_0) \neq 0$, so $\delta \neq 0$; (ii) $\mathrm{rk}\, \Phi(c) \geq r$ on the nonempty Zariski-open set $\{\delta \neq 0\}$, over $\mathbb{Q}$ and $\mathbb{C}$; (iii) if $r = \min(m, n)$ the generic rank is maximal.*

Proof. A matrix over a field has rank $\geq r$ iff some $r \times r$ minor is nonzero, and taking minors commutes with reduction $\mathbb{Z} \to \mathbb{F}_p$, so the nonzero minor mod p is the reduction of an integer $\delta(c_0) \neq 0$. A polynomial with a nonzero value is nonzero, and the coefficient space is irreducible, so $\{\delta \neq 0\}$ is dense open; (iii) adds the bound $\mathrm{rk} \leq \min(m, n)$. □

No good-reduction hypothesis is needed: the certificate uses only "nonzero mod $p \Rightarrow$ nonzero in $\mathbb{Z}$", never the converse, since reduction can only lower rank. The prime is unconstrained and c_0 need not even lie in the smooth locus. Failure mod p is inconclusive and is reported as UNKNOWN, never as a rank deficiency. Finitely many such certificates hold simultaneously on a dense open set of smooth members, because the coefficient space is irreducible. Every certificate records c_0, p, the matrix dimensions and a hash of the matrix, and is replayed by the test suite; the resulting h^q must also satisfy the exact index (3) and Serre duality, which are independent postconditions.

Remark 6.2 (Scope). The certified dimensions hold for the *generic* member of the smooth family with the given configuration. They do not establish the cohomology of a particular member—in particular of a member admitting a chosen free action, whose equations lie in a proper closed subfamily—on which dimensions can jump. For the downstairs analysis we therefore redo the rank computations on a generic member *of the invariant family* (Section 8).

Reproduction. For the published X_{7447} model of [3] the engine reproduces $c_1 = 0$, $\operatorname{ind} V = \operatorname{ind} \wedge^2 V = -12$, $c_2(TX) = c_2(V)$, an exact rational slope witness $t = (5, 29, 15, 11, 5)$, and the generic cohomology $h^\bullet(V) = (0, 12, 0, 0)$, $h^\bullet(\wedge^2 V) = (0, 15, 3, 0)$, with every nontrivial rank certified by Proposition 6.1. Poly-stability witnesses in general are certified exactly: rational t found by lattice search, or real-algebraic t certified by isolating a root and bounding the remaining slopes with rational interval arithmetic; numerical witnesses are recorded as NUMERIC_WITNESS and never accepted.

7 Validation: independent replay and exact catalog agreement

Two kinds of evidence support the searches: an independent implementation that exhausts complete domains without sharing any search logic, and model-for-model agreement with the published catalogs.

7.1 Independent exhaustive replay

A standalone enumerator hard-codes the published topology, re-implements HRR, the subset criterion, the anomaly pairing and the slope obstruction—the latter by Fourier–Motzkin elimination rather than extreme rays—and visits every ordered four-column assignment of the domain, deriving the fifth column, canonicalising, and applying the declared conditions to each distinct class.

geometry	domain	ordered states	valid 5th col.	canonical	surv.	set equality
X_{7862}, $K = 1$	GUT_FULL, pairs ≤ 0	43,046,721	6,765,201	58,781	0	exact
X_{7862}, $K = 1$	GUT_FULL, pairs free	43,046,721	6,765,201	58,781	0	exact
X_{7862}, $K = 1$	GUT_CHIRAL, pairs free	43,046,721	6,765,201	58,781	0	exact
X_{7447}, $K = 1$	GUT_FULL, pairs ≤ 0	3,486,784,401	345,025,251	2,906,223	0	exact

Table 5: Independent replays. The valid-fifth-column counts equal 51^4 and 51^5, matching a closed form (51 of the 81 four-tuples per component have sum in $[-1, 1]$), an arithmetic check on the enumeration itself. The X_{7447} replay took 2,434 s at 783 MiB.

The replay census also shows which condition is load-bearing. On X_{7862}, of the 58,781 canonical states 38,784 pass the subset criterion and 1,932 have the right index; with unrestricted pair indices all 1,932 pass the anomaly pairing and *all* are eliminated by the slope obstruction and by nothing else. On X_{7447} the funnel is 2,906,223 → 2,477,960 → 132,870 → 44,580 → 2,040, and again all 2,040 anomaly survivors fall to the slope obstruction. This is why the slope rule's

soundness and completeness (Section 4.3) matter. The census also confirms $\operatorname{ind} V = \operatorname{ind} \wedge^2 V$ on all 58,781 X_{7862} states, with both always even ($d_X = 2$).

7.2 Exact agreement with the published catalogs

We compare row orbits: survivor sets are canonicalised under summand permutations and permutations of identical ambient factors, and nothing else (dualisation is not an equivalence, as it flips $\operatorname{ind} V$). The published catalog is restricted to the same box. The published database does not state a box per geometry, so "inside the box" is our restriction.

geometry, $\lvert\Gamma\rvert$	K	published	ours, finite test	ours, complete test	set equality
X_{7862}, 2	2	5	5	5	exact
X_{7862}, 2	3	7	8	7	exact (complete)
X_{7862}, 2	4–7	10	11 (at $K = 4$)	10	exact, full catalog
X_{7862}, 4	5	58	–	58	exact, full catalog
X_{7447}, 2	1	0	–	0	exact
X_{7447}, 2	2	56	–	56	exact, predicted
X_{7447}, 2	3	87	–	87	exact, predicted
X_{7447}, 2	4	93	–	93	exact, full catalog, predicted

Table 6: Row-orbit comparison with the published GUT catalog [3, 13]. "Exact" means 0 published-only and 0 ours-only orbits. The X_{7447} counts were predicted from the published $\max|k|$ distribution before each run.

Three points deserve emphasis. First, agreement is *set* equality, not count agreement: an earlier count agreement with Table 4 of [3] at $K = 2$ was upgraded to catalog identity only after retrieving and parsing the database itself (63,325 models on 68 geometries, reproducing the counts stated by its authors). Second, our survivor sets are necessary-condition survivors while the catalog is the poly-stable, anomaly-free, correct-index stage, so agreement means the two condition sets coincide *on these domains*; it is not a proof that our conditions are sufficient in general. Third, the finite slope test fails to reach agreement at $K = 3$ and $K = 4$ because of the single spurious orbit of Example 4.8.

Every one of the 10 published $\{X_{7862}, 2\}$ models, the 58 $\{X_{7862}, 4\}$ models and the 93 $\{X_{7447}, 2\}$ models passes every exact necessary check we implement, including the complete slope obstruction, as soundness requires.

7.3 All 68 favourable CICYs

To test the engine beyond the two benchmarks we ran it, unchanged, on every (geometry, $|\Gamma|$) key of the published GUT catalog: $K = 1$ on all 110 keys (all 68 geometries, $h^{1,1} \leq 6$), $K = 2$ on the 101 keys with $h^{1,1} \leq 5$ and $K = 3$ on the 60 keys with $h^{1,1} \leq 4$, 271 runs in total, each recording the statistics of Table 4. Orbits are taken under summand permutations and under ambient-factor permutations that preserve the configuration matrix up to a relabelling of its columns.

K	$h^{1,1}$	keys	geometries	exact	published in box	recovered	extra orbits
1	2–6	110	68	109	40	**40**	1
2	2–5	101	60	91	9,016	**9,016**	79
3	2–4	60	37	60	342	**342**	0

Table 7: The engine on the whole published catalog. The longest run (X_{7447} with $|\Gamma| = 20$, $K = 2$) took 8,794 s and 947 MiB; every $h^{1,1} = 6$ run at $K = 1$ finished within 484 s.

No published model inside a box is ever missed, on any geometry, and the divisibility

obstruction fires on exactly the keys where the catalog is empty. Our survivor set is a necessary-condition set, and on eleven runs (none of them at $K = 3$) it contains extra orbits—80 in total, ten of the eleven runs on geometries with $d_{ijj} \neq 0$, where the sign test is not exact (Proposition 4.4). Each extra was classified with the tools of Section 4.4:

certified unstable (exact decision, Proposition 4.10)	53
certified *stable* (exact real-algebraic witness)	9
numerically unstable (charge-matrix rank $< h - 1$; not a proof)	18

The 53 are false survivors of the necessary conditions, now excluded exactly; the 18 are very probably the same, but only numerically. Before trusting "certified unstable" we ran the decision on 548 published models, which are poly-stable and must never be rejected: a first implementation rejected 91 of them, which exposed two bugs (a positive root whose isolating interval began at zero was discarded; a polynomial remainder was taken in the wrong order); after fixing both, it rejects none and certifies 214 of them stable with exact witnesses. The check is part of the test suite.

A GUT model missing from the catalog. The nine certified-stable extras are a single model: they share one full triple-intersection fingerprint $\{D(k_a, k_b, k_c),\ c_2 \cdot k_a\}$ up to summand permutation, invariant under any automorphism of the Picard lattice preserving d_{ijk}, across CICYs 5256, 5301 and 5452, which the catalog records as descriptions of one manifold. On CICY 5256 with $|\Gamma| = 4$ its charge vectors are

$$(-2, 0, -1, 1, 0),\ (0, -1, 1, -2, 1),\ (0, 2, 1, 0, -1),\ (1, -2, 1, 0, 0),\ (1, 1, -2, 1, 0).$$

It satisfies $c_1(V) = 0$, the structure-group criterion, $\operatorname{ind} V = \operatorname{ind} \wedge^2 V = -12$, the anomaly and Bogomolov conditions, and it is poly-stable: with $t_5 = 1$, t_4 is the unique root in

$$t_4 \in \left[\tfrac{642595}{64736},\ \tfrac{769226}{77493}\right]$$

of an irreducible degree-12 integer polynomial, the remaining coordinates are polynomials in t_4 proved positive by interval arithmetic, and all five slopes vanish identically modulo that polynomial. The Kähler class has $t_{\max}/t_{\min} \approx 231$. The published catalog lists 2,128 models on this key with charges up to $|k| = 6$, 656 of them in the box $|k| \leq 2$ that contains this model, and none with its fingerprint. The model therefore satisfies every condition the catalog states and is absent from it. We do not know why; a stability search confined to moderate Kähler ratios would miss it. It does not descend with individually equivariant line bundles under any of the four free $\mathbb{Z}_2 \times \mathbb{Z}_2$ actions on this manifold, so it is a new GUT model rather than a new standard model.

Empirical saturation. On X_{7862} with $\Gamma = \mathbb{Z}_2$ the survivor set is literally identical at $K = 4, 5, 6, 7$, and no survivor uses a charge above 4. We record this as EMPIRICAL_SATURATION at $K = 4$. It is not a theorem and no unbounded inference is drawn from it; the slice results of Section 5 are the unbounded statements we can actually prove.

Topology. Recomputing $c_2(TX)$ and the Euler number from the configuration matrices of all 68 favourable CICYs in the catalog reproduces the published values in 68/68 cases each.

8 The Γ-representation on cohomology from first principles

Which states survive the quotient $X \to X/\Gamma$ is decided by the Γ-representation on $H^q(X, L)$. Published databases supply these representations as data. To search rather than verify, we

compute them from the explicit group action: generators acting linearly on the ambient coordinates, together with their action on the defining polynomials, both taken from the published symmetry data [14, 13].

8.1 Characters from the Koszul resolution

The Koszul resolution is Γ-equivariant, so in the representation ring

$$\sum_q (-1)^q \big[H^q(X,L)\big] = \sum_p (-1)^p \sum_q (-1)^q \big[H^q(A, L\otimes \wedge^p N^*)\big]. \tag{8}$$

Each term on the right is elementary. On a factor $\mathbb{P}^n$ on which a group element acts by M, let h_m be the complete homogeneous symmetric function of the eigenvalues of M, computed from the characteristic polynomial by Newton's recursion $h_m = \sum_{i\geq 1}(-1)^{i-1} e_i h_{m-i}$ without extracting eigenvalues. By Bott's theorem and Serre duality

$$\mathrm{tr}\big|_{H^0(\mathbb{P}^n,\mathcal{O}(m))} = h_m \ (m\geq 0), \qquad \mathrm{tr}\big|_{H^n(\mathbb{P}^n,\mathcal{O}(m))} = \det(M)^{m+n}\, h_{-m-n-1} \ (m\leq -n-1),$$

and zero otherwise [15]; Künneth multiplies over factors. The normal bundle carries the action ρ on the defining polynomials: grouping the polynomials by multidegree, the trace on $\wedge^p N^*$ is a product of elementary symmetric functions $e_{p_c}(\rho_c^{-1})$ over the groups. (Ignoring ρ is correct only for invariant polynomials, which most geometries do not have.) For abelian Γ the multiplicity of a character χ is the finite Fourier transform $|\Gamma|^{-1}\sum_g \mathrm{char}(g)\overline{\chi(g)}$ [16]. All arithmetic is exact over cyclotomic fields.

8.2 Separating the degrees

(8) gives only the alternating sum. This suffices for the *chiral* content, $H^1 - H^2$, but not for vector-like pairs: H^1 and H^2 of a pair bundle can be isomorphic as representations and cancel. Two mechanisms separate the degrees.

- *Collapse.* When the bidegree-support test of Section 6 shows the Koszul sequence degenerates at E_1, each $H^q(X,L)$ is the sum of the E_1 characters with $q-p$ equal to that degree.
- *Equivariant map ranks.* Otherwise, when one differential joins two nodes, it is multiplication by the defining polynomial f. For f in the invariant family the differential is Γ-equivariant, so it preserves isotypic components and its rank splits character by character; kernel and cokernel per character give both surviving degrees.

The second mechanism needs the action matrices themselves. On a $\mathbb{P}^1$ factor with block B, the action on $H^0(\mathcal{O}(m))$ is $\mathrm{Sym}^m(B)$ and on $H^1(\mathcal{O}(m))$ it is $\det(B)^{-1}(\mathrm{Sym}^{-m-2}(B))^{-\top}$ by Serre duality; Künneth gives Kronecker products. Their traces agree with the symmetric-function characters—two independent computations of the same numbers—and the equivariance $\Phi\,\rho_{\mathrm{src}}(g) = \rho_{\mathrm{tgt}}(g)\,\Phi$ is *verified* for every element rather than assumed, which tests the action matrices, the parsing of the invariant polynomials and the basis conventions at once. For groups of exponent two all characters are ± 1 and all matrices rational, and the ranks are computed exactly over $\mathbb{Q}$; this makes the largest bundles tractable (ambient dimension 120 in under a second) and agrees with the symbolic computation wherever both were run.

8.3 An equivariance obstruction

Proposition 8.1. *Let $G \subset GL$ be the matrix group generated by a lift of Γ to the ambient coordinates, a central extension of Γ. Then $\mathcal{O}_X(k)$ admits a Γ-equivariant structure compatible with this lift if and only if every central element of G that acts trivially on A acts trivially on*

$\mathcal{O}_A(k)$. *For the* $\mathbb{Z}_2 \times \mathbb{Z}_2$ *action on* X_{7862} *the two generators do not commute in* GL_8*: their commutator is* -1 *on every* $\mathbb{P}^1$ *factor, which is trivial on* A *but acts on* $\mathcal{O}(k)$ *by* $(-1)^{\sum_i k^i}$. *Hence* $\mathcal{O}_X(k)$ *is equivariant exactly when* $\sum_i k^i$ *is even.*

This is checked, not assumed; when it fails the cohomology carries only a projective representation and the computation refuses. A second, independent safeguard rejects any character function whose decomposition is not by nonnegative integers.

8.4 Validation

The published database lists the character tables of every relevant cohomology group. Comparing all four degrees of every table over the 13 geometries that carry downstairs models:

models	2,012
character tables compared (9,181 single, 18,640 pair; all degrees)	27,821
exact agreement	**27,821**

One convention had to be identified. The published tables are the *raw* characters of the lift, and the equivariant structure—χ_a for L_a, $\chi_a+\chi_b$ for L_aL_b—is applied only when the downstairs spectrum is formed; that step is load-bearing, since with it 1,948 of 1,948 published charged spectra are reproduced exactly and without it only 1,503. Only tables that are not invariant under their shift distinguish "raw" from "shifted". There are 694 such tables, all vector-like pair bundles, and all 694 match the raw computation while none matches the shifted one. (An earlier version of our comparison applied the shift to the computed side and reported these 694 as unexplained relabellings; the discrepancy was ours.) A further 640 published pair entries are unevaluated rank expressions and are not compared. Two genuine bugs were found by this comparison: CICY 7487 carries four distinct free $\mathbb{Z}_2 \times \mathbb{Z}_2$ actions and each model must use its own, and the polynomial twist ρ cannot be ignored.

End to end, all 22 published downstairs models on X_{7862} have their complete charged spectrum—chiral and vector-like—derived from first principles and matching the published spectra; every non-collapsing pair bundle involved (21 occurrences of 5 distinct bundles) is resolved by equivariant ranks. The Wilson-line projection itself is exact bookkeeping (Section 9), and against the whole database, with the published characters as input, it reproduces 1,948 of 1,948 complete records.

9 A downstairs search on the tetraquadric quotient

9.1 Wilson lines and equivariant structures

For abelian Γ the Wilson line is a pair of characters: χ_T on the colour triplet and χ_D on the weak doublet of the fundamental $\mathbf{5}$. Everything else follows from $\mathbf{10} = \wedge^2\mathbf{5}$:

multiplet	pieces	Wilson-line character	$6Y$
$\mathbf{10}$	Q, u^c, e^c	$\chi_T\chi_D$, χ_T^2, χ_D^2	$1, -4, 6$
$\overline{\mathbf{5}}$	d^c, L	χ_T^{-1}, χ_D^{-1}	$2, -3$

Lemma 9.1. *The Wilson line lies in* $SU(5)$*, so* $\chi_T^3\chi_D^2 = 1$. *For a group of exponent two this forces* $\chi_T = 1$ *and leaves* χ_D *free.*

A piece with Wilson-line character χ on a bundle with equivariant structure e survives with the multiplicity of the upstairs character $\chi - e$ (in additive notation). Equivariant structures are enumerated subject to $\sum_a e_a = 0$, so that $\det V$ descends. The Wilson line is thus derived, not chosen; for $X_{7862}/(\mathbb{Z}_2 \times \mathbb{Z}_2)$ it reproduces the convention of the published database.

9.2 Results

Starting from the 58 ambient-row orbits of our exhaustive $\Gamma = \mathbb{Z}_2 \times \mathbb{Z}_2$, $K = 5$ scan of X_{7862}, with no published input beyond the symmetry data:

upstairs orbits examined	58
some L_a not equivariant ($\sum_i k^i$ odd, Proposition 8.1)	26
equivariant, all needed cohomology determined	32
no Wilson line and equivariant structure gives an acceptable spectrum	11
three chiral families, no charged chiral exotics, a Higgs pair	**21**

All 10 orbits of the published standard-model database on this quotient are recovered. The other 11 are absent from it. We call them *candidates* and label them A01–A11.

9.3 Audit of the eleven candidates

Each candidate was frozen into an immutable manifest and audited against every condition of the original construction:

condition	result
$c_1(V) = 0$, structure group, anomaly, Bogomolov	11/11
poly-stability with an *exact* witness	11/11 (8 rational, 3 real-algebraic)
$\mathrm{ind}\, V = \mathrm{ind}\, \wedge^2 V = -12$	11/11
$h^\bullet(V) = (0, 12, 0, 0)$, $h^\bullet(\wedge^2 V) = (0, 12 + n_h, n_h, 0)$, $n_h > 0$	11/11
every bundle equivariant; all character tables determined	11/11
three Q, u^c, d^c, L, e^c; no charged chiral exotics; a Higgs pair	11/11
no colour-triplet Higgs	11/11
absent from every published downstairs model	11/11

For charge matrices of rank two the slope-zero locus is a curve, so rationalising a numerical point cannot produce an exact witness; a rational point *on* the curve is found instead (for A02, $t = (15, 15, 3, 1)$). Two of the real-algebraic witnesses have $t_1 = 4 + \sqrt{21}$.

9.4 Why the candidates are absent from the published database

The published standard-model database was built from a GUT scan over $-3 \le k^i_a \le 3$ on $h^{1,1} = 4$ [2, Sec. 5.3], which additionally demanded $h^1(X, L^*_a \otimes L^*_b) < |\Gamma|$ for every pair "to increase the chance that the Higgs triplets can be removed" [1]. We checked both conditions against the data: every published model satisfies them (2,111 of 2,111 with readable cohomology for the second). Every candidate violates at least one:

	$\max \lvert k \rvert$	extra $U(1)$s	n_h	$\max h^1(L^*_a L^*_b)$	configurations	outside the scan by
A01	5	1	3	3	192	charge range
A02	5	2	17	8	336	charge range, h^1 bound
A03	5	1	3	3	192	charge range
A04	4	2	17	8	336	charge range, h^1 bound
A05	4	1	3	3	192	charge range
A06	4	2	17	8	336	charge range, h^1 bound
A07	3	2	17	8	336	h^1 bound
A08	3	1	11	8	192	h^1 bound
A09	3	1	11	8	192	h^1 bound
A10	3	2	17	8	336	h^1 bound
A11	3	2	17	8	336	h^1 bound

So the candidates lie outside the published scan's *declared* domain; they were not missed by it. Two remarks. The h^1 bound is a device to make doublet–triplet splitting likely, not a physical requirement: our computation shows that all eleven nevertheless admit 192–336 equivariant structures and Wilson lines each that keep a Higgs doublet pair and project out every colour triplet. And the six rank-two candidates carry two additional massless $U(1)$s where every published model on this quotient carries one, so they are phenomenologically distinct rather than mere additions to the catalog. A01, A03 and A05 are profile-identical to published models—rank three, one extra $U(1)$, $n_h = 3$—and differ only in charge range.

We stress what the audit does not establish: that these are viable. That is what the Yukawa analysis of the next section addresses.

10 The holomorphic superpotential

10.1 Selection rules as a network-flow problem

Under the unbroken $U(1)^4 = S(U(1)^5)$ the low-energy fields carry charges in $\mathbb{Z}^5/\mathbb{Z}(1,1,1,1,1)$:

$$\mathbf{10}_a : \ e_a, \qquad \overline{\mathbf{5}}_{ab} : \ e_a + e_b, \qquad \mathbf{5}_{ab} : \ -(e_a + e_b), \qquad S_{ab} : \ e_a - e_b,$$

from $H^1(L_a)$, $H^1(L_a \otimes L_b)$, $H^1(L_a^* \otimes L_b^*)$ and $H^1(L_a \otimes L_b^*)$ respectively. An operator is neutral when its total charge $c \in \mathbb{Z}^5$ equals $m(1,\dots,1)$; since singlet charges have entry sum zero, $m = \frac{1}{5}\sum_i c_i$ is forced, and the operator is forbidden outright if that is not an integer. Otherwise the singlets must supply the *residual* $r = m\mathbf{1} - c$.

Proposition 10.1 (Singlet insertions)**.** *Let E be the set of pairs (a,b) for which S_{ab} is present, and D_E the directed graph on $\{1,\dots,5\}$ with an edge $a \to b$ for each $(a,b) \in E$. A residual r with $\sum_i r_i = 0$ is the charge of a monomial in the available singlets if and only if D_E carries a nonnegative flow with net outflow r_a at every node a. Integer and real feasibility coincide, and the minimal number of insertions is the value of a transportation problem with costs equal to shortest-path lengths in D_E.*

Proof. A monomial $\prod S_{ab}^{n_{ab}}$ has charge $\sum n_{ab}(e_a - e_b)$, the divergence of the flow n. The node–arc incidence matrix is totally unimodular [8], so a feasible real flow implies an integer one. With unit edge costs and infinite capacities, a min-cost flow decomposes into paths from sources to sinks, each routed along a shortest path [17]. □

Membership is thus decided exactly and with *no bound on the insertion order*. FORBIDDEN_BY_SELECTION_RULE is an exact vanishing statement to all orders in the singlets; an allowed operator is only charge-allowed, and its value needs the cohomology. For comparison with truncated tables we also enumerate every monomial up to a given degree, pruning with the min-cost-flow value as an admissible lower bound, so the enumeration is complete. Two consequences we use later: a rank bound valid to all orders is the *structural rank* of the allowed pattern of a Yukawa matrix; and the singlets that can take a D-flat vacuum value at vanishing Fayet–Iliopoulos terms are exactly those on directed cycles of D_E, since D-flatness, $\sum_S Q_a(S)|s|^2 = 0$, asks for a nonnegative circulation.

10.2 Reproducing every published operator table

The standard-model database [2, 13] records, for every model, thirteen operator tables as lists of singlet monomials. We rebuilt all of them from the charges, for all 2,012 models with complete records (Table 8).

Several conventions are not stated in the sources. Each was found from a failing comparison, diagnosed, then held fixed for the whole database, and the raw (convention-free) agreement is

table	operator	agreement
`YukawaUp`, `YukawaDown`, `MuTerm`	$\mathbf{10105}_H$, $\mathbf{10\overline{55}}_H$, $\mathbf{5}_H\overline{\mathbf{5}}_H$	6,036 / 6,036 tables
`HiggsMass`	full Higgs mass matrix	18,108 / 18,108 cells
`RParVio`	$\overline{\mathbf{5}}^p\mathbf{5}_H$	6,036 / 6,036 cells
`Dim4Proton`	$\mathbf{10\overline{55}}$	36,216 / 36,216 classes
`Dim5Proton`	$\mathbf{10101}\overline{\mathbf{5}}$	60,360 / 60,360 classes
`SuperPotential`	pure-singlet superpotential	2,012 / 2,012 tables
`DTerms`, `FITerms`	D-terms; FI terms $= \mu_{k_a}(t)$	2,012 / 2,012 each
`GMTerm`	$H_u H_d\, f(S, S^\dagger)$	2,012 / 2,012 tables
`NeutrinoKahler`	$L^p H_d^\dagger f(S, S^\dagger)$	6,036 / 6,036 cells
`Kin5`	$\overline{\mathbf{5}}^{p\dagger}\overline{\mathbf{5}}^q f(S, S^\dagger)$	18,108 / 18,108 cells

Table 8: Every operator table of the published standard-model database, reproduced from the $U(1)^4$ charges alone for all 2,012 complete models. The FI-term agreement also checks our intersection numbers on all 13 geometries. The one flag-level difference in the Yukawa tables is a coupling first allowed at insertion order six, above the published truncation, where both lists are empty.

recorded alongside: superpotential lists stop at total operator degree eight capped at singlet order five, Kähler lists at singlet degree three; the Yukawa matrices use the physical Higgs pair `PhysHiggs`; families are ordered by bundle index, which we confirmed by decoding the per-model `EDownSpec` field; the bare bilinear is omitted from `RParVio`; `Kin5` is indexed $\overline{\mathbf{5}}^{p\dagger}\overline{\mathbf{5}}^q$; Kähler monomials never contain a trivial $|S_{ab}|^2$ factor (with such factors only 94 of 2,012 GM tables agree); `NeutrinoKahler` lists only monomials containing a holomorphic singlet, the paper's right-handed neutrino. Two further conventions are more interesting because they show the tables are not always functions of the charges alone. `Dim5Proton` lists an inconsistent subset of the permutations of its three **10** indices—for one index shape sometimes one ordering of three, sometimes two—so only permutation classes are meaningful; per class, every listed cell equals ours. And `Dim4Proton` omits every entry that involves a matter $\overline{\mathbf{5}}$ from the pair bundle of the physical down Higgs (charge-identical to a Yukawa coupling): all 19,450 allowed cells without such a $\overline{\mathbf{5}}$ are listed and all 164 with one are not.

An internal inconsistency. Because these are charge tables, two families from the same line bundle must give identical rows. Using only the published data—family order from each model's own `EDownSpec`—all 2,012 published up-type tables pass this test, but **954 of 2,007 down-type tables fail it**: they contain a zero row where an identical row is nonzero. Transposing or exchanging labels does not repair it (1,028 failures either way). Entry by entry on X_{7862} we reproduce 198/198 up-type and 184/198 down-type cells; the 14 remaining cells lie in exactly five of these inconsistent tables. Section 10.3 shows that where the cohomology can decide, it contradicts the published presentation.

10.3 Leading-order couplings as triple products

At the Abelian locus a Yukawa coupling is the cup product $H^1(X, L_1)\times H^1(X, L_2)\times H^1(X, L_3) \to H^3(X, \mathcal{O}_X) \cong \mathbb{C}$ with $L_1L_2L_3 = \mathcal{O}_X$ [18, 19]. For a hypersurface $X = \{f = 0\}$ of multidegree q in $A = (\mathbb{P}^1)^4$, the Koszul sequence $0 \to \mathcal{O}_A(k-q) \xrightarrow{f} \mathcal{O}_A(k) \to \mathcal{O}_X(k) \to 0$ splits $H^1(X, \mathcal{O}(k))$ into *level 0* (the image of $H^1(A, \mathcal{O}(k))$) and *level 1* (mapping onto $\ker\big(H^2(A, \mathcal{O}(k-q)) \xrightarrow{f} H^2(A, \mathcal{O}(k))\big)$).

Lemma 10.2 (Level-zero vanishing)**.** *If all three classes are at level 0 the coupling vanishes.*

Proof. Restriction commutes with cup product, so the product is the restriction of a class in $H^3(A, \mathcal{O}_A)$, which is zero by Künneth since $H^{>0}(\mathbb{P}^1, \mathcal{O}) = 0$. □

Proposition 10.3 (Level one)**.** *If exactly one class is at level 1, with image $\beta \in H^2(A, \mathcal{O}(k_3 - q))$, and the others are restrictions of $\alpha_i \in H^1(A, \mathcal{O}(k_i))$, the coupling equals the ambient product $\alpha_1\alpha_2\beta \in H^4(A, \mathcal{O}(-q)) \cong \mathbb{C}$.*

Proof. Since $H^3(A, \mathcal{O}_A) = H^4(A, \mathcal{O}_A) = 0$ the connecting map $\delta : H^3(X, \mathcal{O}_X) \to H^4(A, \mathcal{O}(-q))$ is an isomorphism, and δ is linear over restrictions of ambient classes: $\delta(\alpha|_X \cdot \gamma) = \alpha \cdot \delta\gamma$. □

When two or more classes are at level 1 the connecting map is not multiplicative, the naive product vanishes only for degree reasons, and the true coupling needs a lift through f [20]; such couplings are returned as UNKNOWN. (An early version of our code treated them as zero; an audit showed no recorded coupling was of this type, and a regression test now pins a genuine level-sum-two triple to UNKNOWN.) The ambient product is evaluated on Čech monomials: on each $\mathbb{P}^1$ the three exponents must sum to $(-1, -1)$, the generator of $H^1(\mathbb{P}^1, \mathcal{O}(-2))$.

Complex-structure independence. If the relevant Koszul map is zero because its other end is a zero ambient group, the identification of $H^1(X, L)$ with the ambient group holds for *every* smooth member. This was the case for every one of the 774 bundle slots ever evaluated, so every leading-order matrix reported is constant over the invariant family: there are no rank-drop loci at leading order. Couplings where a Koszul map is nonzero are refused.

Downstairs restriction and a convention check. Each slot is restricted to the isotypic component of its piece's Wilson-line character minus the equivariant structure of its bundle, the convention that reproduces 1,948 of 1,948 published spectra. A blocking check requires every slot's isotypic dimension to equal the number of families the spectrum assigns to that bundle. The trilinear form must be Γ-covariant with the character of $H^4(A, \mathcal{O}(-q))$, $\prod_i \det(B_i)^{-1}$, which is trivial exactly when the action descends to the Calabi–Yau quotient. We verified this element by element and index triple by index triple: **zero violations**. That single test validates the Serre-duality bases, the exponent conventions, the Kronecker ordering and the group lift.

Against the published models. The published tables record only which entries are allowed, so we compare our exact leading-order ranks with the published *structural* rank. For both Y_u and Y_d the ranks agree on 20 of 22 X_{7862} models. The two Y_u differences are Lemma 10.2 at work: charge-allowed couplings whose three classes all sit at level 0, identically zero though no charge argument can see it. The two Y_d differences are internally inconsistent published tables; in one, the cohomology gives a 2×3 block of rank two where the published table shows a zero row, and a rank-two 2×3 matrix has a zero row in no basis. Finally, [2, Sec. 6.2] proves that without equivariant blocks Y_u cannot have rank one; our computed Y_u ranks over the 22 models are 0 or 2 and never 1, recovering that theorem from the cohomology rather than from the group theory.

10.4 The Yukawa sector of the downstairs models

We now apply the engine to all 21 orbits of Section 9—the 11 additional ones and, as a baseline, the 10 published ones—through the identical pipeline, over every accepted configuration (192–336 per orbit) and every choice of physical Higgs pair. Singlet spectra are derived from first principles; validated on the published X_{7862} models they agree exactly for 21 of 22, and the 22nd differs only on two singlets whose published multiplicity is an unevaluated rank expression that our equivariant rank computation evaluates (to zero).

Besides the Yukawa matrices we evaluate every dangerous operator that is a genuine triple product: dimension-four R-parity violation at singlet order zero (its QLd^c, LLe^c and $u^cd^cd^c$ pieces), and the one-singlet terms SH_uH_d and SLH_u. Higher-order terms are decided by the selection rule only. We then classify with logical categories, applied identically to both groups:

orbit	published	choices	top at LO	cap Y_u	cap Y_d	QQQL forbidden	RPV_4 at LO	verdict
A01	–	192	–	0	3	all	–	problematic
A02	–	720	–	0	1, 3	all	288	problematic
A03	–	192	–	0	1	all	–	problematic
A04	–	720	–	3	3	none	288	**unknown**
A05	–	192	–	0	0	all	–	problematic
A06	–	720	–	3	3	none	–	**unknown**
A07	–	720	–	0	1	all	288	problematic
A08	–	192	–	0	0	all	–	problematic
A09	–	192	–	0	0	all	–	problematic
A10	–	720	–	0	0	all	–	problematic
A11	–	720	–	0	0	all	–	problematic
P01	1	192	–	0	0	all	–	problematic
P02	2	192	all	2	0	all	–	problematic
P03	3–6	384	–	0	0, 1	all	–	problematic
P04	7	192	–	0	0	all	–	problematic
P05	8	192	–	0	0	all	–	problematic
P06	9–12	384	–	0, 2	2, 3	none	192	problematic
P07	13	192	–	0	1	none	–	problematic
P08	14	192	–	3	3	none	–	unknown
P09	15–18	384	–	0, 3	0, 1, 3	288	96	unknown
P10	19–22	384	192	1, 2	1, 2	none	96	unknown

Table 9: The downstairs models on $X_{7862}/(\mathbb{Z}_2 \times \mathbb{Z}_2)$. "Choices" counts configurations times physical-Higgs choices; "cap" is the largest rank the selection rule allows at any singlet order (sets list the values over choices); counts are numbers of choices. "Published" gives the published model numbers in each orbit.

STRONG_CANDIDATE requires a nonzero top Yukawa at the Abelian locus, rank three reachable for Y_u and Y_d at some singlet order, and dimension-five proton decay, dimension-four and bilinear R-parity violation forbidden to all orders; PHENOMENOLOGICALLY_PROBLEMATIC requires an unconditional exact obstruction—no up- or no down-type Yukawa at any perturbative order, or dimension-four R-parity violation present at the Abelian locus; everything else is UNKNOWN. An orbit is strong if some choice is, problematic if every choice is.

Table 9 gives the result, and it answers the question posed for the new models. **No orbit on this quotient, published or new, is a strong candidate.** Nine of the eleven new orbits are problematic and two unknown; seven of the ten published orbits are problematic and three unknown. The new orbits fail for the same reasons as the published ones, at a comparable rate. In detail, for the eleven new orbits:

- none has a nonzero top Yukawa at the Abelian locus (two published orbits do); in nine the up-type Yukawa is forbidden to *all* perturbative orders;
- A04 and A06 are the only new orbits whose up-type Yukawa can reach rank three, and for both the rank-three pattern survives restricting to singlets that can take D-flat values;
- dimension-five proton decay is forbidden to all orders in nine of eleven, in every configuration;
- dimension-four R-parity violation is computed nonzero at the Abelian locus in 288 of 720 Higgs choices of A02, A04 and A07, and the one-singlet bilinear coupling SLH_u is nonzero in every choice of A04 and A07;
- in A08, SLH_u is charge-allowed but *vanishes cohomologically* in every configuration: an allowed-but-vanishing dangerous operator.

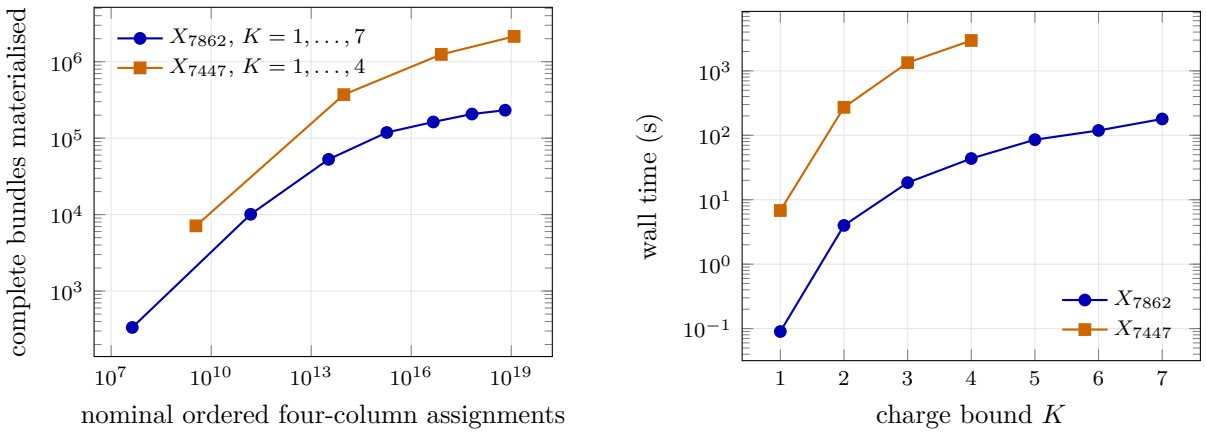


Figure 2: Left: the number of complete bundles the search actually builds against the nominal size of the domain. Across eleven orders of magnitude in the domain the materialised set grows by less than three, because the monotone rules prune entire subtrees. Right: wall time on a single core ($\Gamma = \mathbb{Z}_2$, complete slope rule, bitset-indexed engine; the X_{7862}, $K = 4$ point drops to 28.8 s with the dual slope decision).

The PROBLEMATIC verdicts are exact but perturbative: masses forbidden to all orders by the $U(1)^4$ symmetries can still be generated non-perturbatively where those symmetries are anomalous. The UNKNOWN verdicts rest on the coefficients of higher-order couplings, on the singlet vacuum, and on the normalisation of the physical couplings, none of which is computed here.

11 Performance and scaling

Figure 2 summarises the scaling. Three observations. First, the materialised search space is essentially decoupled from the nominal one: on X_{7862} the nominal domain grows by a factor 1.5×10^{11} from $K = 1$ to $K = 7$ while the bundles built grow by 7×10^2. Second, memory is not the binding resource at this scale: peak resident memory is 189 MiB at X_{7862}, $K = 7$ and 706 MiB at X_{7447}, $K = 4$. Third, the cost of a node falls with depth because of the dual slope decision (Proposition 4.7), so the dominant cost moves from the slope test to enumeration as the search deepens.

Every escalation was preceded by a pilot and a written estimate: a $K = 4$ pilot on 8 roots predicted 110–450 s against an actual 204.8 s (before indexing), and the X_{7447} orbit counts at $K = 2, 3, 4$ were predicted exactly from the published catalog before the runs. For the downstairs stage, computing the Γ-representation on a 120-dimensional ambient cohomology group takes under a second with the exact rational rank path.

The whole test suite—unit, differential (brute force against the split search under all three slope-rule settings), certificate-replay and regression tests—runs in about 30 seconds.

12 Limitations and open problems

What is proved, and in what domain. All UNSAT_BOUNDED results are statements about a declared finite domain, under the declared Kähler chamber T_+ and the declared target. The slice results remove the charge bound only on slices satisfying (7). The only statement with no domain qualifier is the divisibility obstruction. Empirical saturation is not a theorem.

Generic members. All cohomology is that of a Zariski-generic member of the relevant family—the full family for upstairs statements, the Γ-invariant family for downstairs ones. Co-

homology can jump on special members, which can add vector-like pairs or singlets. The leading-order Yukawa matrices are an exception: every coupling evaluated uses Koszul identifications valid for every smooth member.

The Yukawa engine. Triple products are computed for hypersurfaces in products of $\mathbb{P}^1$ and for bundles whose H^1 sits at a single Koszul level. Couplings with Koszul level sum two or more, which need a lift through the defining polynomial, and classes spread over both Koszul levels are returned as UNKNOWN. Neither occurs among the Yukawa couplings of any model studied; both occur for some one-singlet couplings SLH_u of published models, which are therefore reported as undecided. Couplings with singlet insertions are higher products, not triple products, and are decided only by the selection rule. The normalisation of the trace map and the matter Kähler metrics (for which a localisation approach exists [21]) are not computed, so no physical Yukawa coupling or fermion mass is claimed.

Scope of the group theory. Only abelian Γ acting without permuting projective factors is handled; non-abelian groups are refused. The free action and invariant family are input from the published symmetry data.

Published-data discrepancies left open. 954 of 2,007 published down-type Yukawa tables are internally inconsistent as charge tables (Section 10); we report this without reconciling it. (A second apparent discrepancy, in 694 character tables, turned out to be an error in our own comparison convention and is resolved in Section 8.)

Not addressed. Supersymmetry breaking, moduli stabilisation, the full F- and D-flatness of singlet vacua away from vanishing FI terms, and anything non-perturbative. Rules R2 and R3 restrict the domain to that of [3]; searches for models with anti-families or positive pair indices would need them relaxed, which the replays show is supported but costly.

Open problems.

1. *Finiteness on the tetraquadric.* Prove a charge bound on the slices where F is indefinite, by bounding the semialgebraic set $\{F \leq 0,\ \text{ind} \in [-3|\Gamma|, 0]\}$ using the cubic growth of the index along $F \leq 0$ rays; equivalently, settle Conjecture 5.5.

2. *Higher Koszul levels.* Extend the triple-product evaluation to level sum two, and to complete intersections of higher codimension, following the lifting construction of [19, 20].

3. *Singlet vacua.* Decide which singlet vacuum values are F- and D-flat away from the Abelian locus, and hence which of the FULL_RANK_ON_LOCUS verdicts can be realised.

4. *Rank below $h-1$.* Decide exactly the 18 extra survivors whose charge matrix has rank below $h-1$, where the slope-zero locus is positive-dimensional and Proposition 4.10 does not apply; and push the survey of Section 7.3 to larger K, where more models missing from the catalog may appear.

13 Conclusions

We have built a search for heterotic line-bundle models in which every pruning step is a theorem with stated assumptions, and used it to push the analysis from the upstairs GUT stage through the equivariant descent to the holomorphic superpotential.

On the side of *search*, the central observation is that the slope obstruction, used in earlier scans as a finite coefficient test, has an exact and complete form—a fixed polyhedral cone

meeting the span of the charge vectors—that is monotone, prunes every branch, and is decided by a dual test whose cost falls with depth. It removes a spurious candidate the finite test keeps, and with it the search reproduces the published catalogs of both benchmark geometries exactly. The slice decomposition that follows from the same geometry turns 425 slices of the tetraquadric into theorems valid for arbitrarily large charges, and isolates precisely the slices where only the cubic growth of the index can bound the charges. Run over all 68 favourable CICYs, the search never misses a published model, and an exact decision of stability—validated by never rejecting one—shows that the published catalog itself is incomplete: it lacks at least one poly-stable GUT model whose only stability witness lies at a Kähler class with moduli ratio about 1:231.

On the side of *physics*, computing the Γ-representation on cohomology from the group action lets the pipeline discover rather than verify: from our own exhaustive survivor set it recovers all ten published standard-model orbits on the tetraquadric quotient and finds eleven more. Their absence from the published database is explained, in every case, by restrictions the published scan states for itself. The superpotential machinery reproduces every operator table of that database, exposes an internal inconsistency in almost half of its down-type Yukawa tables, and—by evaluating leading-order couplings as genuine triple products—detects vanishings that charge conservation cannot.

The answer to the phenomenological question is sobering and, we think, useful: no model on this quotient, published or new, combines a leading-order top Yukawa with rank-three Yukawa matrices and the absence of dangerous operators. The new models fail in the same ways and at a comparable rate as the published ones. Two of them (A04, A06) remain open, and deciding them requires exactly the ingredients—higher-order couplings, singlet vacua, Kähler metrics—whose absence we have documented rather than papered over.

The methods are not specific to the two benchmarks, as the run over all 68 favourable CICYs shows: the complete slope test needs only the triple intersection numbers, the exact stability decision only the slopes, and the flow formulation of the selection rules is geometry-independent. The natural next steps are that survey at larger charges—where more models missing from the catalog may appear—the exact treatment of rank-deficient charge matrices, the extension of the triple-product evaluation to higher Koszul levels, and the finiteness question of Conjecture 5.5.

A Reproducibility

Everything in this paper is produced by the accompanying repository, which needs only Python and uses exact integer and rational arithmetic throughout; no computer-algebra system or external cohomology package is required. Raw inputs (the published catalogs `GUTall.m` and `sms202.m` and the symmetry data) are stored with SHA-256 checksums. Every search run writes a checkpointed JSON artifact embedding the hash of the source and inputs, and refuses to resume under changed code. The test suite (255 tests: unit, differential, certificate replay and regression tests pinning every number reported here) runs in about 30 seconds.

result	command (from the repository root)
bounded search, e.g. X_{7862}, $K = 3$	`python -m string_sm_sat.cli search --geometry 7862 --K 3 --gamma 2 --slope-rule span`
independent replay (Table 5)	`scripts/independent_replay.py`
catalog comparison (Table 6)	`scripts/compare_catalog.py`
charge-unbounded slices (Section 5)	`scripts/slice_certificate.py`
equivariant cohomology (Section 8)	`scripts/compute_equivariant_cohomology.py`
downstairs search (Section 9)	`scripts/search_standard_models.py`
candidate audit and absence resolution	`scripts/audit_candidates.py`, `scripts/absence_resolution.py`
Yukawa and operator tables (Table 8)	`scripts/yukawa_selection_benchmark.py`, `operator_benchmark.py`, `kahler_benchmark.py`
internal consistency of published tables	`scripts/published_table_consistency.py`
leading-order ranks	`scripts/leading_order_yukawa.py`
singlet-spectrum validation	`scripts/validate_singlet_spectrum.py`
candidate Yukawa sector and ranking (Table 9)	`scripts/candidate_yukawa.py`, `scripts/rank_candidates.py`
68-geometry survey (Table 7)	`scripts/phase8_suite.py`, `phase8_extras.py`
the new GUT model	`scripts/new_gut_model_certificate.py`

The append-only research notebook `RESULTS.md` records every run with its statistics, including corrections to earlier entries; the mathematical notes in `docs/` give the full proofs summarised here.

B Conventions and source discrepancies

Topology. Indices mean $\chi = \sum_q (-1)^q h^q$; J_i are the ordered restricted hyperplane classes; d_{ijl} integrates $J_i J_j J_l$ and $c_{2,i}$ pairs $c_2(TX)$ with J_i. For $c_1(V) = 0$, $c_2(V) = -\frac{1}{2}\sum_a k_a^2$ (Whitney); the positive quadratic expression called $c_2(V)$ in [3, Sec. 5] is $\mathrm{ch}_2(V)$, and we follow the sign of [22, (2.3)], confirmed by the published example.

Published data. Published charge matrices are stored with rows indexed by ambient factors; we transpose to columns. The character tables `GCoh` are indexed by characters in lexicographic label order, with Γ inferred database-wide, and are raw characters of the lift, before the equivariant shift. The equivariant structure enters as a character shift, χ_a for L_a and $\chi_a + \chi_b$ for $L_a L_b$ (fixed by the data: 1,948/1,948 spectra with the shift, 1,503 without). `EDownSpec[[1]]` is [H_u pair, three **10** bundles, H_d pair, three $\overline{\mathbf{5}}$ pairs, singlet codes] with pair index the position among the ten lexicographic pairs and singlet code $2 \cdot (\text{pair index}) + \text{direction}$; `EDownSpec[[2]]` lists the Higgs slots. In the text of [2] $\bar{H}$ denotes the up-type Higgs, while the database field `Subscript["H",a,b]` is the **5** (up-type); both readings are needed to map the Kähler operators correctly.

Discrepancies in the sources, recorded and not silently corrected.

- [4] prints a positive chiral Euler characteristic in one condition while its objective uses the negative target; we follow HRR and the published cohomology.
- [3, Sec. 7] describes the ranks of V and TX as equal (they are five and three); we reproduce c_2 equality only.
- The GUT catalog lists symmetry orders $\{2, 4, 5, 10, 20\}$ for CICY 7447 while the symmetry pages list $\{2, 4, 10\}$; no result depends on it, and order 5 is excluded independently by Proposition 4.12.

- 954 of 2,007 published down-type Yukawa tables are not functions of the charges (Section 10), and the published `Dim5Proton` tables list an inconsistent subset of index permutations.
- The singlet multiplicities of 2,450 entries across 1,197 published records are unevaluated rank expressions; where we evaluate one (CICY 7862, model 2) we find zero at a generic invariant member.

Acknowledgements

[To be completed.] The computations use only exact integer and rational arithmetic and open-source Python; no computer-algebra system or external cohomology package is required.

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